

A Sub-1 ppm/°C Bandgap Voltage Reference With High-Order Temperature Compensation in 0.18- μ m CMOS Process

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Abstract—This paper presents a high-precision bandgap voltage reference (BGR) with high-order temperature compensation. The compensation signal is generated by using both strong-inversion MOSFETs and Bipolar Junction transistors (BJTs), which cancels the high-order nonlinear term $T \ln(T)$ in the BJT base-emitter voltage (V_{BE}), and thus a low temperature coefficient (TC) over a wide temperature range is achieved. The proposed BGR circuit is fabricated in a 0.18- μ m CMOS process with an active area of 0.256 mm² and a max power consumption of 1.35 mW. A minimum TC of 0.706 ppm/°C from −25 °C to 125 °C is achieved after an 8-bit resistance trimming. The line sensitivity is 0.0146%/V operating from 3.2 V to 3.7 V. The BGR achieves a power supply rejection (PSR) of −63.4 dB and a noise spectrum density of 0.92 μ V/ $\sqrt{\text{Hz}}$ at 10 Hz.

Index Terms—Bandgap voltage reference, high-order temperature compensation, temperature coefficient.

I. INTRODUCTION

PRECISE voltage references, which can provide stable operating points, play an indispensable role in electronic systems [1], [2]. Since it was introduced by Widlar [3], bandgap voltage reference (BGR) technology has been widely used to generate on-chip reference voltage. The output voltage of BGR is related to the bandgap energy of semiconductor materials [4]. In bulk CMOS, BJTs are often used in BGR design, whose base-emitter voltage V_{BE} is complementary-to-absolute-temperature (CTAT), and can be compensated with a proportional-to-absolute-temperature (PTAT) voltage ΔV_{BE} to form a temperature-insensitive voltage [5]. MOSFETs can also be used in voltage reference design by utilizing their temperature-sensitive threshold voltage, which is however

more prone to process-spread [6]–[10]. Besides temperature-insensitivity, care must be taken to overcome the impacts of supply voltage and process variations induced output shifts. Introducing trimming techniques (such as digital control word or laser cut) can resist these adverse changes. To achieve a lower temperature sensitivity, power consumption and area are often compromised [11], [12].

The temperature coefficients (TCs) of first-order temperature-compensated BJT-based BGRs are typically in the range of 20 ~ 100 ppm/°C [13]–[16]. To maintain high accuracy in a wide temperature range, higher-order compensation circuits are used to cancel the curvature content in V_{BE} . Many temperature compensation techniques to suppress temperature-dependent nonlinearity [11], [17]–[25] have been reported. [11] and [17] are based on switched capacitors. In [11], a temperature-independent current was required, which may introduce additional errors due to process variations. In [18] and [19], the exponential behavior of weak-inversion-region MOSFETs are utilized to compensate for the $T \ln(T)$ nonlinearity. Piecewise compensation was reported in [20]–[24]. Reference [20] achieves a low area of 0.0094 mm² and a low current consumption of 28 μ A with a 1.67 ppm/°C minimum TC. In [23], a five-piece linear compensation was implemented with a minimum measured TC of 0.9 ppm/°C and an average measured TC of 3 ppm/°C. Reference [25] is based on temperature-dependent resistor ratio, whose underlying problem is that the variations of resistor formed by different materials are not the same during manufacturing, thus reducing the BGR accuracy. In [26], V_{G0r} , the extrapolated bandgap voltage at 0 K, is extracted from the temperature characteristic of V_{BE} to compensate the nonlinear temperature dependence and the minimum measured TC was 0.8 ppm/°C. It is difficult to ensure the accuracy of the output voltage if the high-order temperature compensation is realized without resistors [27], therefore the resistive trimming network is still common. In most of the literature, TCs are in the single-digit ppm/°C range since the nonlinearity is not completely canceled [23], thus achieving a sub-1 ppm/°C TC is still a great challenge in a standard CMOS process.

In this paper, a high-precision BGR with high-order temperature compensation scheme is presented. Two BJTs and two MOSFETs biased in their strong-inversion regions are

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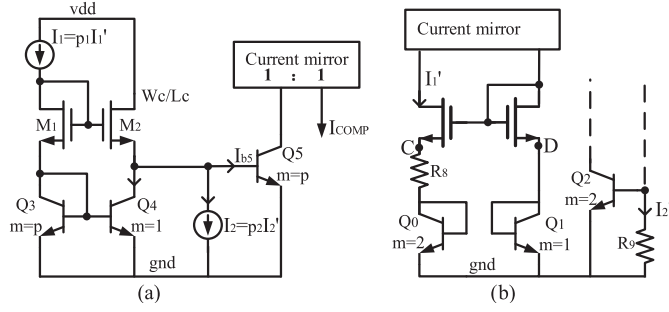


Fig. 1. (a) Simplified topology of the proposed high-order compensation circuit (b) Corresponding bias circuit.

combined to generate a higher-order compensation signal. This signal, expressed in the analytical form of $T \exp(f(T))$, can be designed close to the $T \ln(T)$ term existing in V_{BE} , thus non-linearity compensation can be achieved. In this work, binary-coded resistor networks are used to trim TC and the output voltage amplitude. The proposed BGR circuit is fabricated in a $0.18\text{-}\mu\text{m}$ CMOS process, and the measurement results prove that a voltage reference with a low TC of sub-1 ppm/°C can be obtained, validating our compensation method.

The rest of this paper is organized as follows. Section II analyzes the proposed high-order compensation method. Section III presents the circuit implementation. Section IV shows the measurement results, followed by a conclusion in Section V.

II. PROPOSED HIGH-ORDER TEMPERATURE COMPENSATION SCHEME

In most bandgap topologies [1]–[4], BJT in diode configuration (the base connected to the collector) behaves as the main component, that is the voltage V_{BE} of the diode-configured BJT provides the bandgap-energy-related component. According to previous studies, the thermal dependence of V_{BE} can be expressed as [28]

$$V_{BE}(T) = V_{GO} - [V_{GO} - V_{BE0}(T_r)] \frac{T}{T_r} - (\eta - \theta) V_T \ln\left(\frac{T}{T_r}\right) \quad (1)$$

where V_{GO} is the extrapolated bandgap voltage at 0°K , T_r is the temperature where the reference's TC is zero, usually chosen to be near room temperature [11], $V_{BE0}(T_r)$ is the base-emitter voltage at T_r , η is a positive constant related to the process whose typical value is 3~4 [28], θ is the temperature exponent of the collector bias current. V_T is the thermal voltage and equals kT/q , where k is the Boltzmann constant, q is the electron charge. As in (1), V_{BE} contains a linear term and a high-order term $V_T \ln(T/T_r)$.

The nonlinear term of V_{BE} is ignored in first-order BGR, but it has to be considered when low TC is required. In this work, a compensation circuit shown in Fig. 1(a) is proposed to achieve high-order $T \ln(T)$ compensation. In the following, its operation principle is analyzed theoretically with its effectiveness verified with Spectre simulation.

The bias currents used in the compensation circuit, I_1 and I_2 , are generated by the circuit in Fig. 1(b). A simple

operational amplifier is used to force the voltage of node C and node D equal. So the voltage across R_8 is the base-emitter voltage difference of Q_0 and Q_1 , expressed as $V_T \ln(I_{C1} I_{S0} / (I_{S1} I_{C0}))$, where I_C is the collector current, and I_S is the saturation current. The emitter area of Q_0 is twice that of Q_1 . Therefore, the voltage across R_8 can be expressed as $(k \ln 2 / q) T$. The voltage across R_9 is the base-emitter voltage of Q_2 . Then, they are mirrored to get the bias currents, I_1 and I_2 , which are expressed as

$$\begin{cases} I_1 = p_1 \frac{k \ln 2}{q R_8} T \\ I_2 = p_2 \frac{V_{BE2}}{R_9} \end{cases} \quad (2)$$

where p_1 and p_2 are mirror ratios.

In Fig. 1(b), M_1 and M_2 have identical dimensions and they are biased in their strong inversion regions. BJTs, Q_3, Q_4 and Q_5 , are biased in their forward regions. The drain current I_{DS} of a MOSFET working in its strong inversion region is expressed as

$$I_{DS}(T) = \frac{\mu(T) C_{ox} \frac{W}{L}}{2} (V_{GS}(T) - V_{TH}(T))^2 \quad (3)$$

where C_{ox} is the gate capacitance per unit area, W and L correspond to the channel width and length, respectively. Carrier mobility $\mu(T)$ and threshold voltage $V_{TH}(T)$ are both temperature-dependent and expressed as [4]

$$\begin{cases} V_{TH}(T) = V_{TH0} - \alpha(T - T_r) \\ \mu(T) = \mu_0 \left(\frac{T}{T_r}\right)^{-m} \end{cases} \quad (4)$$

where V_{TH0} is the threshold voltage at T_r and α is a positive constant. μ_0 is the carrier mobility at T_r , and m is a positive constant. The nominal value of temperature order m in $\mu(T)$ is 1.5 [4].

By applying KVL in Fig. 1(a), $V_{GS1} + V_{BE3} = V_{GS2} + V_{BE5}$ holds. For the sake of simplicity, ΔV_{BE} is used to replace the base-emitter voltage difference between Q_5 and Q_3 and ΔV_{GS} is used to replace the gate-source voltage difference between M_1 and M_2 , so it is rewrote as

$$\Delta V_{GS} = \Delta V_{BE} \quad (5)$$

Since the emitter area of Q_3 is the same as that of Q_5 , ΔV_{BE} is expressed as

$$\Delta V_{BE} = V_T \ln\left(\frac{I_{C5}}{I_{C3}}\right) = V_T \ln\left(\frac{I_{COMP}}{I_1}\right) \quad (6)$$

Q_3 and Q_4 form a current mirror, so the drain current of M_2 can be estimated as $I_{DS2} = \frac{1}{p} I_1 + I_2 + I_{b5}$. The V_{SB} of M_1 and M_2 are approximately equal, so it can be considered that M_1 and M_2 have the same threshold voltages. ΔV_{GS} is expressed as

$$\Delta V_{GS} = \sqrt{\frac{2}{\mu(T) C_{ox} \frac{W_c}{L_c}}} (\sqrt{I_1} - \sqrt{\frac{1}{p} I_1 + I_2 + \frac{I_{COMP}}{\beta(T)}}) \quad (7)$$

After substituting (6), (7) into (5)

$$V_T \ln\left(\frac{I_{COMP}}{I_1}\right) = \sqrt{\frac{2}{\mu(T)C_{ox}\frac{W_C}{L_C}}} \cdot \left(\sqrt{I_1} - \sqrt{\frac{1}{p}I_1 + I_2 + \frac{I_{COMP}}{\beta(T)}}\right) \quad (8)$$

$\beta(T)$ is the common-emitter current gain of BJTs and it can be expressed as [29] $\beta(T) = \beta_\infty \exp(-\frac{\Delta E_G}{kT})$, where ΔE_G is the bandgap narrowing factor of emitter [22], which is proportional to the BJT emitter doping concentration, β_∞ is the maximum common-emitter current gain of BJT, which is temperature-independent.

It is a transcendental equation and is difficult to get the exact form of I_{COMP} . During design, the base current of Q_5 , $I_{COMP}/\beta(T)$, is designed much smaller than $\frac{1}{p}I_1 + I_2$, and it can be ignored in (8). As a result, the expression of the target compensation current is derived as

$$I_{COMP} = I_1 \exp\left(\frac{\sqrt{\frac{2}{\mu(T)C_{ox}\frac{W_C}{L_C}}}(\sqrt{I_1} - \sqrt{\frac{1}{p}I_1 + I_2})}{V_T}\right) \quad (9)$$

Replace I_1 and I_2 with (2)

$$\begin{aligned} I_{COMP} &= p_1 \frac{k \ln 2}{q R_8} T \\ &\cdot \exp\left(\frac{\sqrt{\frac{2}{\mu(T)C_{ox}\frac{W_C}{L_C}}}(\sqrt{p_1 \frac{k \ln 2}{q R_8} T} - \sqrt{\frac{p_1 \frac{k \ln 2}{q R_8} T + p_2 \frac{V_{BE2}}{R_9}}})}{kT/q}\right) \end{aligned} \quad (10)$$

It shows that I_{COMP} is highly temperature-dependent and nonlinear. The purpose of high-order temperature-compensated BGR is to reduce the $T \ln(T/T_r)$ term in V_{BE} as described in (1). Our derived current I_{COMP} can be designed (with the help of numerical tools like Matlab) to match the $T \ln(T)$ term to achieve compensation. The values of R_8 and R_9 are first determined coarsely. The coefficient of T^m , that depends on C_{OX} , $\frac{W_C}{L_C}$ and $\mu_0 T_r^m$, is estimated by the simulated saturation current and simulated V_{GS} of M_1 , while V_{BE2} is estimated as a linear function of temperature. p_1 , p_2 are the current-mirror gains and they can be simply set to be 0.5 or a value that is thought appropriate. In this design, 0.5 is set considering the power. p is the emitter area ratio of the BJTs, with usual values of 1, 2, 4, 8, 24, and so on, considering the matching in the layout. And the optimal value of p for the best TC can be determined by sweeping its values. In this design, p_1 , p_2 and p are finally set to 0.5, 0.5, 4, respectively. Compared to $T \ln(T)$, Fig. 2 shows that (10) is a close estimation of $T \ln(T)$, with our selected device sizing and device ratios ($R_8 = 2.49 \text{ k}\Omega$, $R_9 = 39.53 \text{ k}\Omega$, $W_C/L_C = 2.4\mu/2\mu$). For comparison, the unit of T is degree Celsius. If the unit of T is Kelvin, the curve of $T \ln(T)$ will add a linear term related to T to approach the analytical curve of (10), and the added linear term can be compensated in the BGR core circuit described in Section III. In order to facilitate analysis and visualization, the data are normalized. Fig. 2 shows

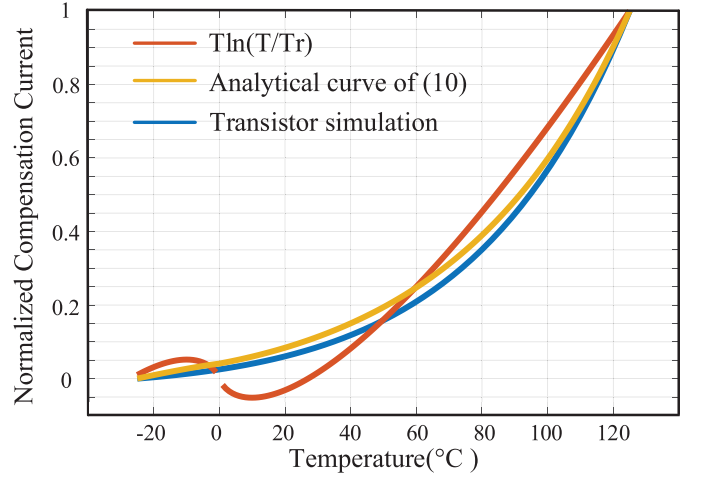


Fig. 2. Normalized curve for $T \ln(T/T_r)$, the analytical form of I_{COMP} in (10) using estimated process parameter, and the transistor simulation result of I_{COMP} .

the comparison between $T \ln(T/T_r)$ and theoretical curve. The error is around 5% over the whole temperature range, except when the value of $T \ln(T/T_r)$ is negative. Fig. 2 also shows the circuit simulation result of I_{COMP} generated using the proposed circuit in Fig. 1. Good agreement between the simplified analytical model and simulation can be achieved. The parameters in (10) are estimated and it will introduce errors in the actual situation. And m equals to 1.5 which lead to the plurality and makes the analysis more complicated. When designing the circuit, there is a lack of accurate process parameter model in the simulator, causing the deviation observed in the post-layout measurement. Therefore, in order to solve these problems, it is necessary to trim the networks.

III. PROPOSED BGR IMPLEMENTATION

The proposed BGR is detailed in Fig. 3, in which the start-up circuit is omitted for simplicity. This BGR consists of a bias current generator, a high-order compensation current generator, and a voltage reference core. The currents I_1 and I_2 in high-order temperature compensation circuit are not only used to generate the compensation current I_{COMP} , but also bias the operational amplifier (A_1). In order to avoid the interaction between the two subcircuits, the PTAT currents, i.e., I_1 and I_3 , are generated separately. In the bias generator, a cascode current mirror is adopted to improve the PSR without burning extra power.

A. Bias Current Generation

A cascode current mirror and a pair of NMOSFETs are used to force node C and D to have the same potential. How to generate I_1 and I_2 has been discussed in Section II. They are mirrored to bias A_1 and the transistors M_1 and M_2 working in the strong inversion region. And the mirror ratios, p_1 and p_2 , in (2) depend on the transistor aspect ratios.

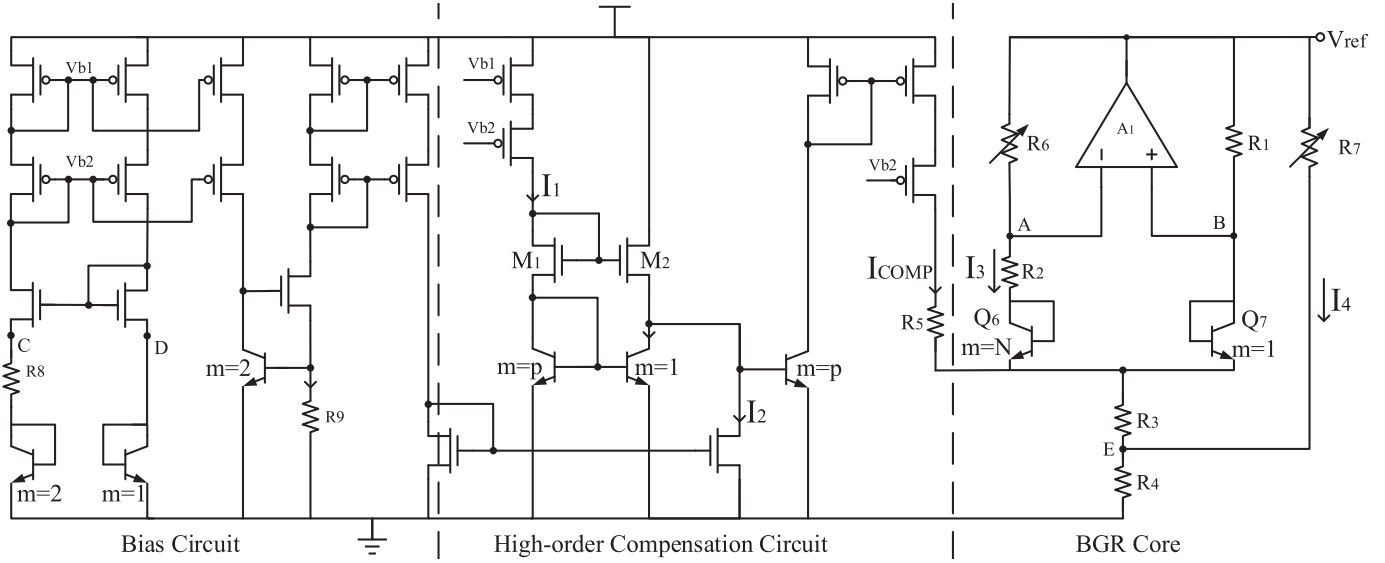


Fig. 3. Schematic of the proposed BGR structure, omitting the start-up circuit for simplicity.

B. High-Order Compensation Current Generation and V_{ref} Generation

The principle of the employed first-order temperature-compensated bandgap is conventional, except for introducing a scaling factor. To improve PSR, the output of A_1 and one terminal of R_1 , R_6 , R_7 are coupled to the output instead of supply. A well-known phenomenon is that the classical value of the bandgap output V_{BGR} is about 1.2 V. However, the BGR designed is required to provide a larger reference voltage. So R_3 , R_4 and R_7 are utilized to generate a scaling factor over 1. A scaling factor smaller than 1 is also feasible, depending on the application requirements [30].

Biased in the forward region, Q_6 and Q_7 , form the reference core. A_1 follows the cascode architecture with gain boosting to offer large DC gain of 130 dB. A_1 and the feedback circuit force the node A and B to have the same potential, so the PTAT voltage is produced by the V_{BE} difference between Q_6 and Q_7 . The PTAT current is generated across R_2 , expressed as $I_3(T) = (V_{BE7} - V_{BE6})/R_2 = (kT \ln N)/(qR_2)$, where $N=8$ is the emitter area ratio of Q_6 and Q_7 . The current I_4 injected to the tail resistor R_4 is expressed as

$$I_4 = \frac{V_{ref} - (2I_3 + I_{COMP})R_4}{R_4 + R_7} \quad (11)$$

According to the KVL derivation in the BGR core, the voltage V_{ref} can be expressed as (12).

$$V_{ref} = \left(1 + \frac{R_4}{R_7}\right) [V_{BE6} + (R_6 + R_2 + 2R_3 + \frac{2R_4}{1 + R_4/R_7})I_3 + (R_3 + \frac{R_4}{1 + R_4/R_7})I_{COMP}] \quad (12)$$

It can be seen that R_6 mainly affects the linear compensation term, and R_7 affects the amplitude, the linear compensation term and the higher-order compensation term. R_7 in this design is mainly used to trim the amplitude of V_{ref} against PVT variation, which is not mandatory in a normal BGR. Substitute I_3 into (12), the final expression of V_{ref} is in (13). It can be seen that the coefficient of amplitude, linear compensation

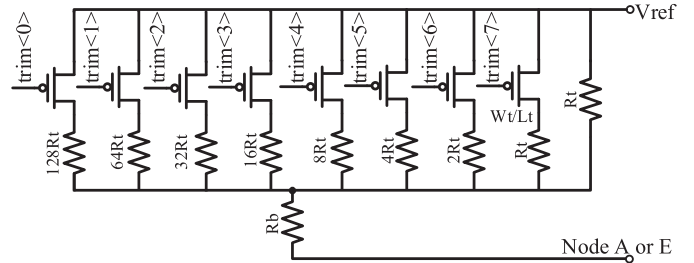


Fig. 4. Trimming network topology consisted of paralleled resistors.

term and high-order compensation term are all resistor ratios, while the compensation term D is not, whose PVT influence can be trimmed as discussed below.

$$\begin{cases} V_{ref} = (1 + \frac{R_4}{R_7})[V_{BE6} + aT + bc \cdot \exp(d)T] \\ a = \frac{R_6 + R_2 + 2R_3 + 2R_4/(1 + R_4/R_7) k \ln N}{R_2 q} \\ b = \frac{R_3 + R_4/(1 + R_4/R_7)}{R_8} \\ c = \frac{p_1 k \ln 2}{q} \\ d = \frac{\sqrt{\frac{2}{\mu(T)C_{ox} \frac{W_c}{L_c}}} (\sqrt{\frac{c}{R_8} T} - \sqrt{\frac{c}{pR_8} T + p_2 \frac{V_{BE2}}{R_9}})}{kT/q} \end{cases} \quad (13)$$

C. Resistive Trimming Network

Trimming is a typical method to suppress process and chip package influence [6], [35]. As aforementioned, the linear variations (proportional to T) of V_{ref} can be suppressed by mainly adjusting R_6 , while the amplitude of V_{ref} can be adjusted by mainly tuning R_7 . Fig. 4 shows the 8-bit binary-weighted resistor trimming network used in this work. Large MOSFET ($19.8\mu/600n$) are used for the resistor switches to minimize their on-resistance influence. R_6 and R_7 employ the same topology but different nominal resistance, with R_t and R_b being 2.36 k Ω , 56.18 k Ω , and 2.38 k Ω , 5.11 k Ω ,

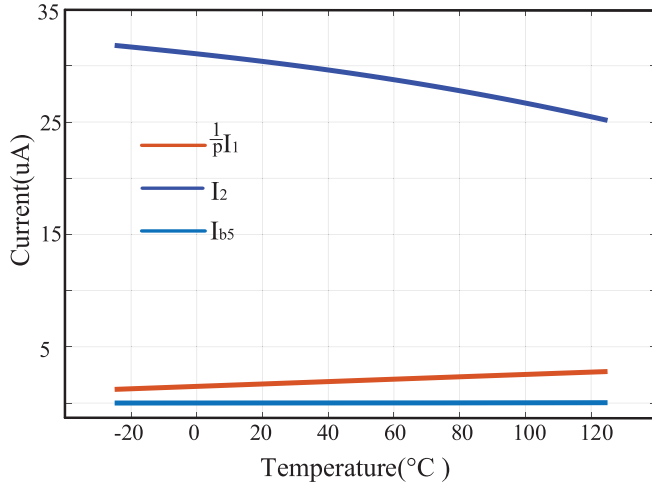
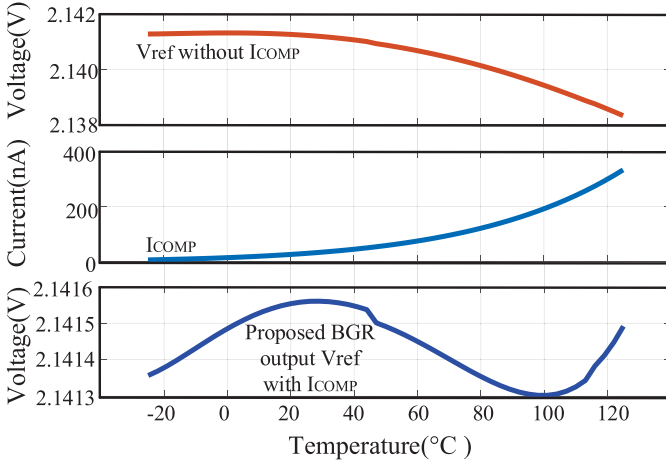
Fig. 5. Simulated currents of $\frac{1}{p}I_1$, I_2 and I_{b5} .

Fig. 6. Simulated temperature dependence of the proposed BGR.

TABLE I
DESIGN PARAMETERS

Parameter	Value	Parameter	Value
R_1	58.54 k Ω	R_2	7.05 k Ω
R_3	3.29 k Ω	R_4	5.11 k Ω
R_5	543 Ω	R_8	2.49 k Ω
R_9	39.53 k Ω	N:1	8:1
R_t for R_6	2.36 k Ω	R_b for R_6	56.18 k Ω
R_t for R_7	2.38 k Ω	R_b for R_7	5.11 k Ω
W_t/L_t	19.8 μ /600n	W_c/L_c	2.4 μ /2 μ

respectively. The binary weighted resistor ranges from 33.42% to 100% of R_t , making the tuning range very wide. As a result, the output (2.031 V $\sim$ 2.284 V) can be adjusted to 2.14 V with good stability over temperature.

D. Circuit Simulation

Fig. 5 shows the simulated currents of $\frac{1}{p}I_1$, I_2 and I_{b5} . It can be seen that at 27 °C, $\frac{1}{p}I_1$ is 1.77 μ A, I_2 is 30.13 μ A, and I_{b5} is 6.19 nA. It is a good evidence that I_{b5} can be ignored, as assumed in Section II. Fig. 6 shows BGR output as a function of temperature, showing a TC 9.3 $\text{ppm}/^\circ\text{C}$ before and an greatly improved TC of 0.611 $\text{ppm}/^\circ\text{C}$ after applying our proposed high-order compensation. For more fair comparison, Fig. 7 compares our result with the fully

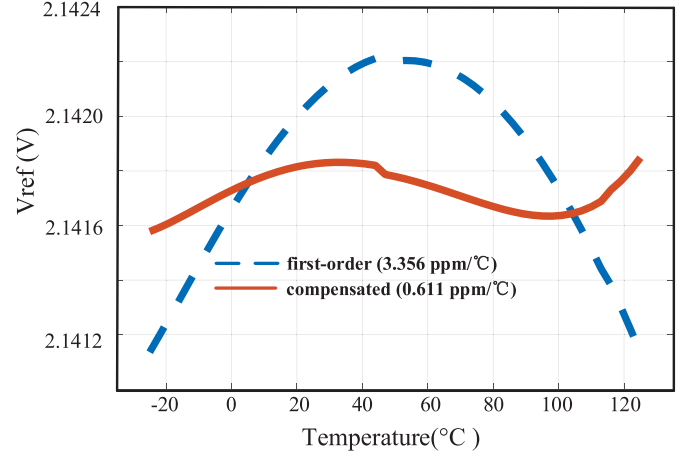


Fig. 7. TC performance comparison between the optimized first-order BGR and the proposed BGR with high-order compensation.

TABLE II
POST-LAYOUT SIMULATED TC UNDER DIFFERENT
TRANSISTOR AND RESISTOR CORNERS

Transistor Corner	Resistor Corner	TC ($\text{ppm}/^\circ\text{C}$)
tt	tt	0.611
	ff	1.221
	ss	0.658
ff	tt	1.448
	ff	0.673
	ss	1.827
ss	tt	1.506
	ff	1.870
	ss	1.185
sf	tt	0.666
	ff	1.192
	ss	0.955
fs	tt	0.878
	ff	1.354
	ss	0.911

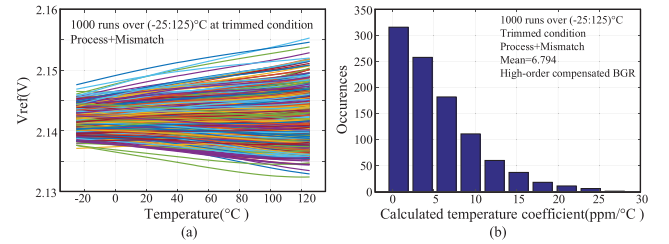


Fig. 8. (a) Monte Carlo simulation results and (b) the histogram of the calculated TC from (a).

first-order compensated output, indicating the effectiveness of I_{COMP} for high-order correction. The nominal values of main parameters in the circuit is listed in Table I.

Table II summarizes the post-layout simulation TC results for MOS transistors and resistors corners. The control words are trimmed at each different corners. The best achieved TC after trimming is 0.611 $\text{ppm}/^\circ\text{C}$ and the worst is 1.870 $\text{ppm}/^\circ\text{C}$, demonstrating the robustness of this design. To verify the effects of mismatch and process, Monte Carlo simulation of 1000 runs is performed, including all the components (NPNs, resistors, MOS transistors and capacitors). The Monte Carlo simulation of the trimmed high-order BGR was carried out using a fixed control word which is obtained in tt corners. Fig. 8(a) presents V_{ref} as a function of

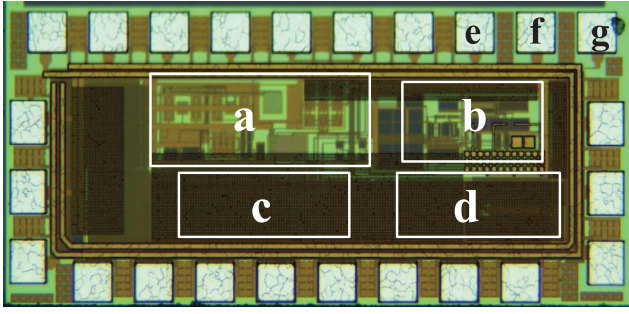


Fig. 9. Chip microphotograph with an active area of $908 \mu\text{m} \times 282 \mu\text{m}$. a: High-order Compensation Circuit; b: Operational Amplifier; c: Amplitude trimming; d: TC trimming; e, f and g are test pads.

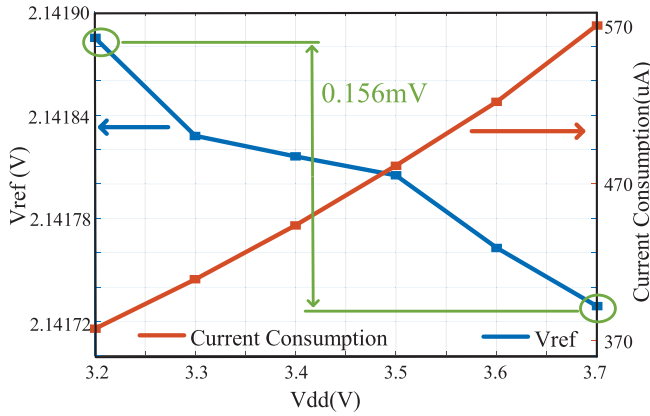


Fig. 10. Measured output voltage and current consumption at room temperature as the function of supply voltage.

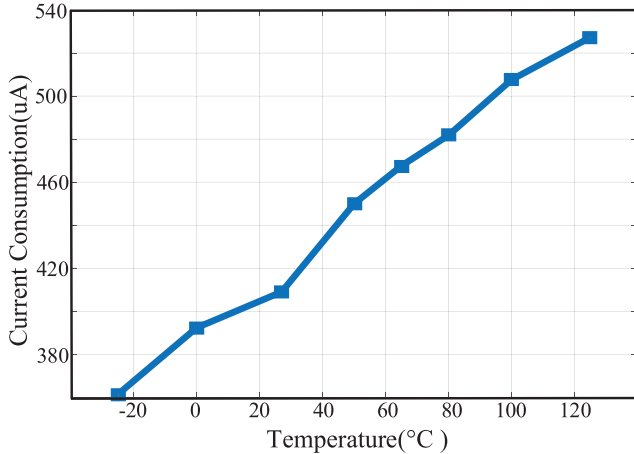


Fig. 11. Measured current consumption as a function of temperature.

temperature, with its TC distributions showing in Fig. 8(b). Across 1000 runs, an average TC of $6.794 \text{ ppm}/^\circ\text{C}$ is achieved. Sigma is not meaningful for non-Gaussian distribution [4], [19]. With the help of the resistor trimming network, 13.3% of samples show $\text{TC} > 12 \text{ ppm}/^\circ\text{C}$.

IV. MEASUREMENT RESULTS

The proposed bandgap reference was manufactured in a $0.18\text{-}\mu\text{m}$ mixed-signal CMOS process with an area of 0.256 mm^2 . N-type polysilicon resistors were used to implement all the resistors. Fig. 9 shows the chip micro-graph.

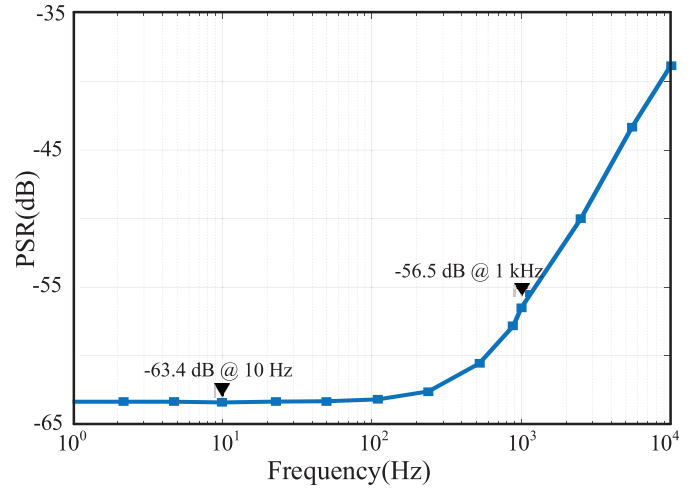


Fig. 12. Measured results of PSR at room temperature with 3.3 V supply.

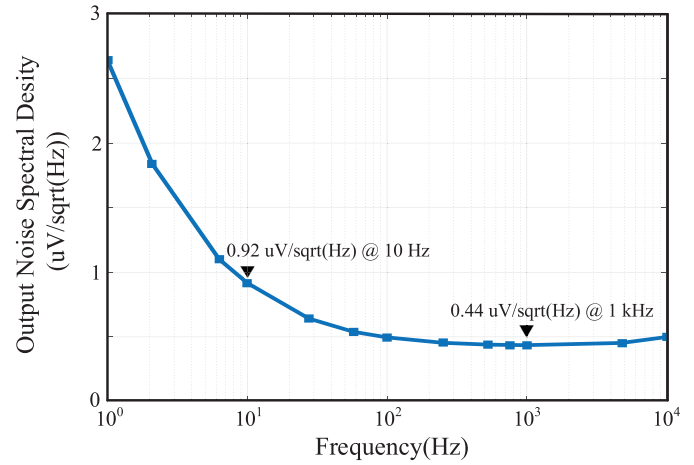


Fig. 13. Measured results of the output noise voltage spectral density.

TABLE III
MEASURED TC OF 6 SAMPLES

sample	TC before trimming $\text{ppm}/^\circ\text{C}$	TC after trimming $\text{ppm}/^\circ\text{C}$
1	7.4	0.706
2	15.2	0.803
3	13.6	1.245
4	8.7	0.98
5	12.5	1.186
6	20.3	1.535

The BGR is directly wire-bonded to the PCB for in-lab characterization.

Fig. 10 presents the measured output voltage and current consumption at room temperature with supply voltage varying from 3.2 V to 3.7 V, corresponding to a line sensitivity of $0.0146\%/V$. Its current consumption increases from $377 \mu\text{A}$ to $571 \mu\text{A}$, mainly for the reason that the dc operating point of A_1 is drifted when VDD has been changed. The power consumption difference can be reduced by improving the PSR of A_1 . Fig. 11 shows the measured current consumption, which increases from $361 \mu\text{A}$ to $527 \mu\text{A}$ from -25°C to 125°C , respectively. The measured PSR of this design is shown in Fig. 12, showing a rejection of -63.4 dB at 10 Hz and -56.5 dB at 1 kHz. Fig. 13 shows the measured noise

TABLE IV
PERFORMANCE SUMMARY AND COMPARISON TABLE

	[26] TCASI'21	[32] JSSC'21	[33] JSSC'21	[31] TCASI'20	[23] TCASI'19	[27] TCASI'19	This work
Tech.(nm)	65 CMOS	180 CMOS	130 CMOS	65 CMOS	350 CMOS	350 CMOS	180 CMOS
Supply Voltage (V)	2.5~3.6	1.8	2~3.3	0.5~1	2.6~5	2~5	3.2~3.7
Line Sensitivity (%/V)	NA	NA	0.03	0.64	0.0041	0.175	0.0146
Current Consumption(μA)	40+	17	120	0.072	94	33	409
PSR (dB)	NA	-76@DC	-82@10Hz -30@100kHz	-50@DC	-83@DC	-61@100Hz -57@10kHz	-63.4@10Hz -56.5@1kHz
Noise ($V/\sqrt{Hz}$)	NA	NA	NA	NA	1.2 μ @1Hz	NA	0.92μ@10Hz 0.44μ@1kHz
Integrated Noise	10 μV_{pp} 0.1Hz-10Hz	56 μV_{rms} *	84.3 μV_{rms} 0.1Hz-10Hz ^o	NA	3.6 μV_{rms} 0.1Hz-10Hz	NA	15 μV_{rms} 1Hz-1kHz
Temp. Range ($^{\circ}C$)	0~80	-40~125	-40~150	-40~120	-45~125	-40~125	-25~125
TC ($ppm/^{\circ}C$)	0.8(min) 0.87(max)	3.2(min) 5.5(max) 4.3(avg)	5.78(min) 13.5(max) 8.75(avg)	38(min) 70(max) 42(avg)	0.9(min) 7.52(max) 3(avg)	1.01(min) [^] 3.88(max) [^]	0.7(min) 1.557(max) 1.183(avg)
Curvature Correction	Yes	Yes	Yes	No	Yes	Yes	Yes
# of samples	3	18	7	6	10	10 [^]	6
Active Area (mm^2)	0.006+	0.38	0.08	0.0522	0.0616	0.0396	0.256
FoM ($^{\circ}C^3/(W \times mm^2)$)	13.3+	0.73	0.20	358.49	2.13	10.31	0.093

+ : Power and area of off-chip op-amps are not included. Five commercial op-amps used here consume 7.3 mA typical current in total. So the power, area, and FoM comparison with it are not meaningful.

*: It is inferred by the measured integrated noise referred to the input of the $\Delta\Sigma$ ADC.

^o: 84.3 μV_{rms} is the integrated output noise up to 10 Hz before offset cancellation enabled, and after it enabled, the integrated output noise is 175.5 μV_{rms} .

[^]: The optimized TC in [27] is from 1.01 $ppm/^{\circ}C$ to 3.88 $ppm/^{\circ}C$ in which 10 references are respectively trimmed. While for 30 samples after trimming, the mean value of TC is 4.03 $ppm/^{\circ}C$ with a standard deviation of 2.31 $ppm/^{\circ}C$.

NA: No Information Available

spectrum without output loading. It reveals that the noise is 0.92 $\mu V/\sqrt{Hz}$ at 10 Hz and 0.44 $\mu V/\sqrt{Hz}$ at 1 kHz. The corner frequency is about 100 Hz. The integrated noise is 15 μV_{rms} from 1 Hz to 1 kHz.

For TC measurements, six chips were tested. During the test, an insulated cover is used to protect the chip to avoid magnetic field interference from the external environment. For the measurement setup, OYO1000 Programmable Temperature Chamber (PTC) is used for temperature sweep, whose temperature range is from $-70^{\circ}C$ to $150^{\circ}C$. The voltage is measured by Keysight 34401A Digital Multimeter with 1 μV resolution. Twenty minutes of soak time at each temperature point is used for temperature stabilization. The averaged output of 300 voltage readings is used to minimize noise-induced error. For trimming and TC calculation, firstly, R_7 is adjusted to make the output amplitude close to 2.14 V at room temperature and then R_7 is fixed. Secondly, the optimum values of R_6 are measured for both $-25^{\circ}C$ and $100^{\circ}C$ and R_6 is then fixed by averaging those two optimum values. After the trimming procedure, the chips are verified again from $-25^{\circ}C$ to $125^{\circ}C$ with 11 temperature steps and the TC can be finally obtained. Table III shows the measured TC before/after trimming of 6 samples for the temperature from $-25^{\circ}C$ to $125^{\circ}C$. Their measured output voltage versus temperature is shown in Fig. 14. To avoid invalid output measurements during temperature settling procedure, longer soak time can help to further smooth the TC curvature. It can be observed from the measured results that there is a significant improvement in terms of temperature stability after the digital trimming using control words. The difference between the maximum

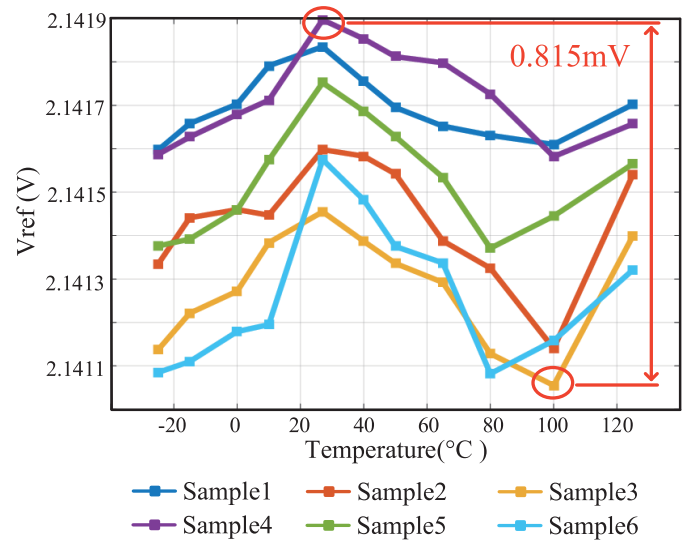


Fig. 14. Measured output voltage versus temperature of 6 samples.

voltage and the minimum voltage exhibited by any of these references between $-25^{\circ}C$ and $125^{\circ}C$ is 0.815 mV peak to peak. The proposed bandgap exhibits a minimum measured TC performance of 0.706 $ppm/^{\circ}C$, and the maximum one of 6 samples is 1.535 $ppm/^{\circ}C$. The average measured TC is 1.076 $ppm/^{\circ}C$, making the proposed BGR with digital trimming competitive.

Table IV summarizes the performance of the proposed BGR in comparison to other reported results. Compared with [31]–[33], the proposed BGR obtains a better TC; while it

obtains a better noise performance than [32] and a better line sensitivity than [31], [33]. The proposed BGR obtains a comparable TC with [26], [27]; while it obtains a wider temperature range than [26] and a better line sensitivity than [27]. The proposed BGR obtains a comparable TC and noise performance with [23]; while it obtains a better average TC. The area and power consumption of this proposed BGR is larger than the other works mainly due to A_1 in this work. There are no special requirements for the operational amplifier. Therefore, a basic 5-transistor operational amplifier or with a lower power consumption like [34] can be used.

V. CONCLUSION

The proposed high-precision BGR with high-order temperature compensation has been manufactured in 0.18 μm CMOS technology. The combination of MOSFETs working in the strong inversion region and BJTs generates the high-order compensation current to cancel out the nonlinear term in V_{BE} . The proposed topology achieves a minimum measured performance of 0.706 ppm/°C in the temperature range of -25°C to 125°C with a 3.3 V supply and 1.35 mW power consumption at room temperature. The fact that the proposed BGR has excellent stability over a wide temperature range makes it a good choice to supply high-performance applications such as high-resolution ADCs and high-performance voltage regulators, etc.

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